

STAT6203: Theoretical Statistics III

Robust Statistics

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1. Location estimation

Consider the problem of estimating an unknown location parameter ξ from a sample

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F(\cdot - \xi).$$

In general, we search for suitably *efficient* estimators. Huber (1964) seeks to obtain *robust* (in a certain sense) estimators which perform well when F is only being approximately known; in particular, suppose that F is a slightly ‘contaminated’ version of a known distribution F_0 .

Definition 1.1 (Tukey-Huber Contamination). Given $\varepsilon \in [0, 1]$ and a distribution F_0 , define

$$\mathcal{P}_\varepsilon(F_0) := \{(1 - \varepsilon)F_0 + \varepsilon H : \text{arbitrary distribution } H\}.$$

Remark. $\mathcal{P}_\varepsilon(F_0)$ is contained within the ε -Kolmogorov-Smirnov neighborhood of F_0 : for any $F = (1 - \varepsilon)F_0 + \varepsilon H$,

$$\|F - F_0\|_\infty = \|(1 - \varepsilon)F_0 + \varepsilon H - F_0\|_\infty = \varepsilon\|H - F_0\|_\infty \leq \varepsilon.$$

With this, we may examine how an estimator based for a parameter of F_0 suffers under the contaminated sample from $F \in \mathcal{P}_\varepsilon(F_0)$. As a first step, we may look at some simple population-level functionals.

Example 1.2. Consider the mean functional $\mu(G) := \mathbb{E}_G[X]$. The ‘bias’ of μ on $F := (1 - \varepsilon)F_0 + \varepsilon H$ against F_0 is

$$\mu(F) - \mu(F_0) = \varepsilon\mu(H) - \varepsilon\mu(F_0).$$

Note that this is completely unrestricted over $F \in \mathcal{P}_\varepsilon(F_0)$ even if we fix ε , owing to the arbitrary nature of H (which in principle may not even have a mean!).

Example 1.3. Consider the (left) median functional $m(G) := G^-(\frac{1}{2}) = \inf\{x : G(x) \geq \frac{1}{2}\}$, and let $\varepsilon < \frac{1}{2}$. For $F := (1 - \varepsilon)F_0 + \varepsilon H$, note that

$$m(F) = \inf\left\{x : (1 - \varepsilon)F_0(x) + \varepsilon H(x) \geq \frac{1}{2}\right\}.$$

Since $0 \leq H(x) \leq 1$, we can bound

$$\begin{aligned} F_0^-\left(\frac{\frac{1}{2} - \varepsilon}{1 - \varepsilon}\right) &= \inf\left\{x : (1 - \varepsilon)F_0(x) + \varepsilon \geq \frac{1}{2}\right\} \\ &\leq m(F) \\ &\leq \inf\left\{x : (1 - \varepsilon)F_0(x) \geq \frac{1}{2}\right\} = F_0^-\left(\frac{\frac{1}{2}}{1 - \varepsilon}\right). \end{aligned}$$

Unlike the previous example, the median $m(F)$ cannot deviate arbitrarily over $F \in \mathcal{P}_\varepsilon(F_0)$.

For the purposes of location estimation, it is natural to restrict ourselves to *translation invariant* estimators.

Definition 1.4 (Translation Invariant Estimators). We say that an estimator T is translation invariant if $T(X + a\mathbf{1}) = T(X) + a$ for all samples X and $a \in \mathbb{R}$. We denote the class of all translation invariant estimators by $\mathcal{T}$.

Definition 1.5 (Asymptotic Bias). Let $\{T_n\}$ be a sequence of estimators for $T(F_0)$. Its asymptotic bias is defined as

$$b(\{T_n\}, F) := \left| \lim_{n \rightarrow \infty} \mathbb{E}_F[T_n] - T(F_0) \right|.$$

We can now choose our ‘best’ estimator on the basis of asymptotic bias; fix $\varepsilon > 0$ and consider the minimax problem

$$\min_{\{T_n\} \in \mathcal{T}} \max_{F \in \mathcal{P}_\varepsilon(F_0)} b(\{T_n\}, F).$$

Bibliography

Huber, P. J. (1964). Robust Estimation of a Location Parameter. *The Annals of Mathematical Statistics*, 35(1), 73–101. <https://doi.org/10.1214/aoms/1177703732>